

Numerical Analysis

Lab 4

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Direct methods for linear systems I

This lab deals with the numerical resolution of a linear system

$$\mathbb{A}\mathbf{x} = \mathbf{b}, \quad \text{for } \mathbb{A} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n.$$

Today we will focus on direct methods (i.e. methods for which the solution is obtained in a **finite number** of operations).

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Backward and forward substitution methods I

Consider a generic non singular lower triangular system

$$\mathbb{L}\mathbf{x} = \mathbf{b}$$

In order to have $\mathbb{L}$ non singular, its diagonal elements must be non zero.
The system reads;

$$\begin{bmatrix} l_{1,1} & 0 & \cdots & 0 \\ l_{2,1} & l_{2,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n,1} & l_{n,2} & \cdots & l_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Forward substitution

$$x_1 = \frac{b_1}{l_{11}},$$
$$x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j=1}^{i-1} l_{ij} x_j \right), \quad i = 2, \dots, n.$$

Comments

Computational cost: $\mathcal{O}(n^2)$.

Backward and forward substitution methods III

Similarly, consider a generic non singular upper triangular system

$$\mathbb{U}\mathbf{x} = \mathbf{b}$$

Again, in order to have $\mathbb{U}$ non singular, its diagonal elements must be non zero. The system reads;

$$\begin{bmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,n} \\ 0 & u_{2,2} & \cdots & u_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Backward substitution

$$x_n = \frac{b_n}{u_{nn}},$$
$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij} x_j \right), \quad i = n-1, \dots, 1.$$

Comments

Computational cost: $\mathcal{O}(n^2)$.

Exercise 4.1

- a Write a function which implements the backward substitution method for solving a generic upper triangular system. Apply such a function to solve the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

- b Write a function to implement the forward substitution method to solve lower triangular systems. Apply this function to the system $\mathbb{A}^T \mathbf{x} = \mathbf{b}$, with $\mathbb{A}$ and $\mathbf{b}$ defined as above.
- c Use the backward and the forward algorithms to solve the system $\mathbb{A}^T \mathbb{A} \mathbf{x} = \mathbf{b}$.

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1 Solution of triangular systems

2 LU decomposition

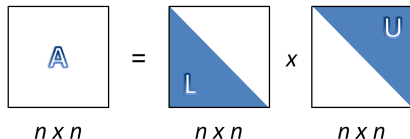
3 Cholesky decomposition

LU decomposition I

Let $\mathbb{A} \in \mathbb{R}^{n \times n}$: the LU decomposition of $\mathbb{A}$ allows to write

$$\mathbb{A} = \mathbb{L}\mathbb{U}$$

with $\mathbb{L}$ **lower triangular** matrix s.t. $l_{ii} = 1, \forall i = 1 \dots n$, and $\mathbb{U}$ **upper triangular** matrix.



The existence and the uniqueness of such decomposition is related to the principal submatrices of order i of the matrix $\mathbb{A}$, with $i = 1, \dots, n-1$, which are demanded to be non singular (necessary and sufficient condition). As an alternative, the sufficient condition holds:

- $\mathbb{A}$ is a strictly **diagonally dominant** by row $|a_{ii}| \geq \sum_{j=1, j \neq i}^n |a_{ij}| \forall i$ OR by column matrix
- $\mathbb{A}$ is a **symmetric positive definite** matrix. $\mathbb{A} = \mathbb{A}^T$ and $\mathbf{x}^T \mathbb{A} \mathbf{x} > 0 \quad \forall \mathbf{x} \neq \mathbf{0}$.

LU decomposition in MATLAB/Octave

```
>> help lu
lu    lu factorization
[L,U,P] = lu(A)
returns unit lower triangular matrix L, upper
triangular matrix U, and permutation matrix P so that
P*A = L*U.
```



```
[L,U] = lu(A)
stores an upper triangular matrix in U and a "
psychologically lower triangular matrix" (i.e. the
product of a lower triangular and a permutation matrix
) in L, so that A = L*U, A can be rectangular.
```

LU decomposition I

LU decomposition with no pivoting

$$\left. \begin{array}{l} \mathbf{A}\mathbf{x} = \mathbf{b} \\ \mathbf{A} = \mathbf{L}\mathbf{U} \end{array} \right\} \longrightarrow \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b} \longrightarrow \left\{ \begin{array}{l} \mathbf{L}\mathbf{y} = \mathbf{b} \\ \mathbf{U}\mathbf{x} = \mathbf{y} \end{array} \right.$$

When the standard LU factorization fails, dealing with non-singular matrices, we can resort to the **LU factorization with pivoting**.

Pivoting is a technique which changes the order of the rows of a matrix in order to avoid null pivot elements during the LU factorization.

LU decomposition with pivoting

$$\left. \begin{array}{l} \mathbf{A}\mathbf{x} = \mathbf{b} \\ \mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U} \end{array} \right\} \longrightarrow \mathbf{P}\mathbf{A}\mathbf{x} = \mathbf{b} \longrightarrow \left\{ \begin{array}{l} \mathbf{L}\mathbf{y} = \mathbf{P}\mathbf{b} \\ \mathbf{U}\mathbf{x} = \mathbf{y} \end{array} \right.$$

Notice that the right-hand side of the lower triangular system is modified with respect to the standard LU factorization

LU decomposition I

This approach is ideal to solve multiple systems $\mathbb{A}\mathbf{x} = \mathbf{b}_i$ with different right hand-sides $\mathbf{b}_i$, and all sharing the same coefficient matrix. Actually it suffices to compute the decomposition $\mathbb{LU}$ just **once**.

Exercise 4.2

Consider the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 2 & 10 & 4 & 0 \\ 1 & 0 & 2 & 2 \\ 1 & 4 & 0 & 2 \\ 1 & 2 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 10 \\ 1 \\ 3 \\ 3 \end{bmatrix}.$$

- a Compute the solution of the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ using the $\mathbb{LU}$ decomposition of matrix $\mathbb{A}$.
- b Find the determinant of $\mathbb{A}$ without using the command `det`.

Exercise 4.3

Let us consider the matrix $\mathbb{A} = \begin{bmatrix} 50 & 1 & 3 \\ 1 & 6 & 0 \\ 3 & 0 & 1 \end{bmatrix}$, $\mathbb{B} = \begin{bmatrix} 4 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & 15 \end{bmatrix}$.

- 1 Apply the Matlab[®] command `lu` and compute the LU factorization of the matrices $\mathbb{A}$, $\mathbb{B}$.
- 2 Compute the determinant of the $\mathbb{A}, \mathbb{B}$ without the command `det` if possible.
- 3 Solve the systems $\mathbb{A}\mathbf{x} = \mathbf{a}, \mathbb{B}\mathbf{x} = \mathbf{b}$. Choose the vectors $\mathbf{a}$, $\mathbf{b}$, such that the solution of the system is $\mathbf{x}_{ex} = [1, 1, 1]^T$.
- 4 Compute the solution of the system $\mathbb{A}\mathbf{x} = \mathbf{b}$, by using the backward and forward substitution the functions previously implemented.

Exercise 4.4 (LU decomposition)

Consider the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 12 & 5 & -8 & -5 \\ -4 & -4 & 8 & -6 \\ 4 & 2 & -3 & 0 \\ 0 & -1 & 2 & -4 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 4 \\ -6 \\ 3 \\ -3 \end{bmatrix}.$$

- a Use the LU decomposition of matrix $\mathbb{A}$ to solve the system. Use the `lu` MATLAB/Octave command.
- b Compute the determinant of $\mathbb{A}$ using the LU decomposition.
- c Consider now a different right-hand side vector $\mathbf{c} = [-22, -12, -1, -11]^T$. Use an approach which allow us to contain the computational cost.

Inverse Matrix

We can define the inverse of a squared matrix $\mathbb{A} \in \mathbb{R}^{n \times n}$ as the matrix $\mathbb{X} = \mathbb{A}^{-1} \in \mathbb{R}^{n \times n}$ such that $\mathbb{A}\mathbb{X} = \mathbb{X}\mathbb{A} = \mathbb{I}$.

It is possible to determine $\mathbb{A}^{-1}$ by solving the following n linear systems:

$$\mathbb{A}\mathbf{v}_i = \mathbf{e}_i, \quad i = 1, \dots, n,$$

where $\mathbf{e}_i$ denote the consecutive columns of the matrix $\mathbb{I}$ (i.e the vectors of the standard basis of $\mathbb{R}^n$). Thus it turns out that

$$\mathbb{A}^{-1} = [\mathbf{v}_1 | \mathbf{v}_2 | \dots | \mathbf{v}_n]$$

Exercise 4.5

- 1 Write a function `[InvA]=MyInv(A)` that computes the inverse `InvA` of a generic square matrix `A`.
- 2 Use the function `MyInv` to compute the inverse of the matrix $\mathbb{A}$ in the previous exercise. Compare the result with the output provided by Matlab with command `inv`.

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Cholesky decomposition I

Let $\mathbb{A} \in \mathbb{R}^{n \times n}$ be a **symmetric** and **positive definite** matrix: then there $\exists!$ **upper triangular**, with $h_{ii} > 0$, s.t.

$$\mathbb{A} = \mathbb{H}^T \mathbb{H}.$$

For any symmetric matrix $\mathbb{A}$

- | | | |
|------------------------|--------|--|
| | | <ul style="list-style-type: none">• $\mathbf{x}^T \mathbb{A} \mathbf{x} > 0, \quad \forall \mathbf{x} \neq \mathbf{0}$ |
| $\mathbb{A}$ pos. def. | $\iff$ | <ul style="list-style-type: none">• $\lambda_i > 0, \quad \forall i = 1, \dots, n, \quad \lambda_i$ eigenvalue of A• $\det(\mathbb{A}_i) > 0, \quad \forall i = 1, \dots, n$ |
| $\mathbb{A}$ pos. def. | $\iff$ | <ul style="list-style-type: none">• $\mathbb{A}$ is diagonally dominant, with all the diagonal elements being positive. |

Cholesky decomposition II

The elements of $\mathbb{H}$ can be computed by the following algorithm:

Cholesky decomposition

We set $h_{11} = \sqrt{a_{11}}$ and for $i = 2, \dots, n$,

$$h_{ij} = \frac{1}{h_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} h_{ik} h_{jk} \right) \quad j = 1, \dots, i-1$$

$$h_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} h_{ik}^2}$$

Cholesky decomposition III

Cholesky factorization is available in MATLAB via the command $R = \text{chol}(A)$, where R is the upper triangular matrix $\mathbb{H}$.

Cholesky decomposition in MATLAB

```
>> help chol
```

```
CHOL    Cholesky factorization.
```

```
CHOL(A) uses only the diagonal and upper triangle of A. The lower triangle is assumed to be the (complex conjugate) transpose of the upper triangle. If A is positive definite, then  $R = \text{CHOL}(A)$  produces an upper triangular  $R$  so that  $R' * R = A$ . If A is not positive definite, an error message is printed.
```

Attention! `chol` command **does not** check the symmetry of the matrix!

Cholesky decomposition IV

LU decomposition with no pivoting

$$\left. \begin{array}{l} \mathbb{A}\mathbf{x} = \mathbf{b} \\ \mathbb{A} = \mathbb{L}\mathbb{U} \end{array} \right\} \longrightarrow \mathbb{L}\mathbb{U}\mathbf{x} = \mathbf{b} \longrightarrow \left\{ \begin{array}{l} \mathbb{L}\mathbf{y} = \mathbf{b} \\ \mathbb{U}\mathbf{x} = \mathbf{y} \end{array} \right.$$

Cholesky decomposition

$$\left. \begin{array}{l} \mathbb{A}\mathbf{x} = \mathbf{b} \\ \mathbb{A} = \mathbb{H}^T\mathbb{H} \end{array} \right\} \longrightarrow \mathbb{H}^T\mathbb{H}\mathbf{x} = \mathbf{b} \longrightarrow \left\{ \begin{array}{l} \mathbb{H}^T\mathbf{y} = \mathbf{b} \\ \mathbb{H}\mathbf{x} = \mathbf{y} \end{array} \right.$$

When you have to solve more systems $\mathbb{A}\mathbf{x} = \mathbf{b}$ with **varying $\mathbf{b}$** and $\mathbb{A}$ s.p.d., compute just **once** the decomposition.

Computational cost: $\mathcal{O}(\frac{2}{3}n^3)$ for LU decomposition, $\mathcal{O}(\frac{1}{3}n^3)$ for Cholesky decomposition.

Exercise 4.6

Consider the matrix

$$\mathbb{A} = \begin{bmatrix} 44 & 15 & 29 & 26 & 119 \\ 15 & 33 & 32 & 18 & 15 \\ 29 & 32 & 252 & 112 & 73 \\ 26 & 18 & 112 & 124 & 90 \\ 119 & 15 & 73 & 90 & 430 \end{bmatrix}$$

- 1 Verify that the Cholesky decomposition can be applied to matrix $\mathbb{A}$ (hint: use the command `eig`);
- 2 Write a Matlab function with signature `[H] = MyChol(A)` to implement the Cholesky decomposition of the matrix $\mathbb{A}$;
- 3 Compute the Cholesky decomposition of the matrix $\mathbb{A}$ by means of both the Matlab command `chol` and the function `MyChol` and compare the results;
- 4 Solve the linear system with $\mathbb{A}\mathbf{x} = \mathbf{b}$, $\mathbf{b} = [1, 1, 1, 1, 1]^T$

Exercise 4.7

Consider the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 10 & 0 & 3 & 0 \\ 0 & 5 & 0 & -2 \\ 3 & 0 & 5 & 0 \\ 0 & -2 & 0 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}.$$

- a Compute the solution of the system. Use the Cholesky decomposition of matrix $\mathbb{A}$ if possible (check the hypotheses).
- b Find the determinant of $\mathbb{A}$ without using the Matlab command `det`.

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4 Homework

Homework 4.1

Consider the linear system $\mathbb{E}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{E} = \begin{bmatrix} 4 & 1 & 1 & 1 & 5 \\ 4 & 1 & 2 & 0 & 0 \\ 1 & 0 & 15 & 5 & 1 \\ 0 & 2 & 4 & 10 & 2 \\ 3 & 1 & 2 & 4 & 20 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 12 \\ 19 \\ 22 \\ 18 \\ 30 \end{bmatrix}$$

- 1 Verify with Matlab that the sufficient conditions for the existence and uniqueness of the LU decomposition without pivoting are not fulfilled by matrix $\mathbb{E}$.
- 2 Compute the LU of matrix $\mathbb{E}$ with the Matlab command `lu` and verify if pivoting takes place with the Matlab command `spy`.
- 3 Solve the system $\mathbb{E}\mathbf{x} = \mathbf{b}$ by using the LU decomposition at the previous item.

Homework 4.2

Consider the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 1 & 4 & 0 & 2 \\ 2 & 10 & 4 & 0 \\ 1 & 0 & 2 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ 10 \\ 1 \end{bmatrix}.$$

- a Compute the solution of the system $\mathbb{A}\mathbf{x} = \mathbf{b}$ using the LU decomposition with pivoting of matrix $\mathbb{A}$.
- b Find the determinant of $\mathbb{A}$ (use the Matlab command `det` only to compute the determinant of $\mathbb{P}$).

Homework 4.3

Consider the tridiagonal matrix $\mathbb{A}_n \in \mathbb{R}^{n \times n}$ defined as

$$\mathbb{A}_n = \begin{bmatrix} a & b & & & \\ b & a & b & & \\ & \ddots & \ddots & \ddots & \\ & & b & a & b \\ & & & b & a \end{bmatrix}$$

for $a = 2$ and $b = 1$.

- a Verify with MATLAB that $\mathbb{A}_{10}$ is a symmetric positive definite matrix.
- b Provide the form of the matrix $\mathbb{V}_{10}$ such that $\mathbb{A}_{10} = \mathbb{V}_{10}^T \mathbb{V}_{10}$

References I